

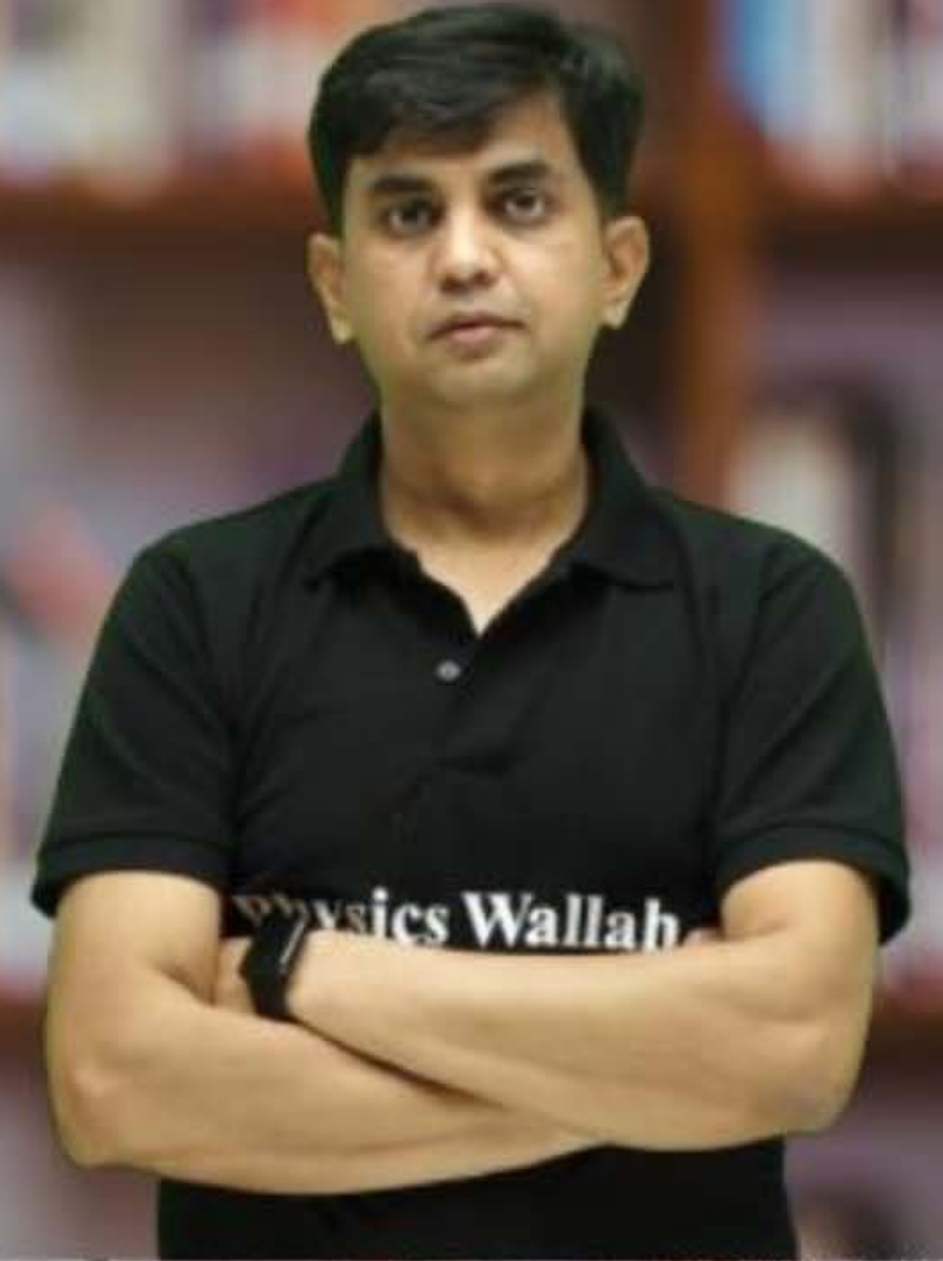
CS & IT ENGINEERING



Computer Network

TCP Connection Close

Lecture No. - 02



By - Abhishek Sir



Recap of Previous Lecture



Topic

TCP Connection Establishment





Topics to be Covered



Topic

TCP Connection Close



ABOUT ME

Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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Topic : TCP Operation



Three phases of TCP operation :

1. Connection establishment

→ (3-way handshake) (SYN)

2. Data transfer

3. Connection termination

→ (4-way handshake) (FIN)



Topic : TCP Connection Establishment

→ Initial Sequence Number : if SYN flag is on

[both TCP client and server randomly chooses their initial sequence number, to prevent from some kind of attacks]

* ISN $\Rightarrow$ 32-bit (seq. no.)
 $\searrow$ first seq. no

P₁

TCP Client

P₂

TCP Server

Client_ISN = X

Seq_no = X, ACK_no = Garbage
 SYN_flag = 1, ACK_flag = 0
 (No Payload)

Client_ISN = X

TCP Client

TCP Server

Seq_no = X, ACK_no = Garbage
 SYN_flag = 1, ACK_flag = 0

Seq_no = , ACK_no = (X+1)
 SYN_flag = , ACK_flag = 1

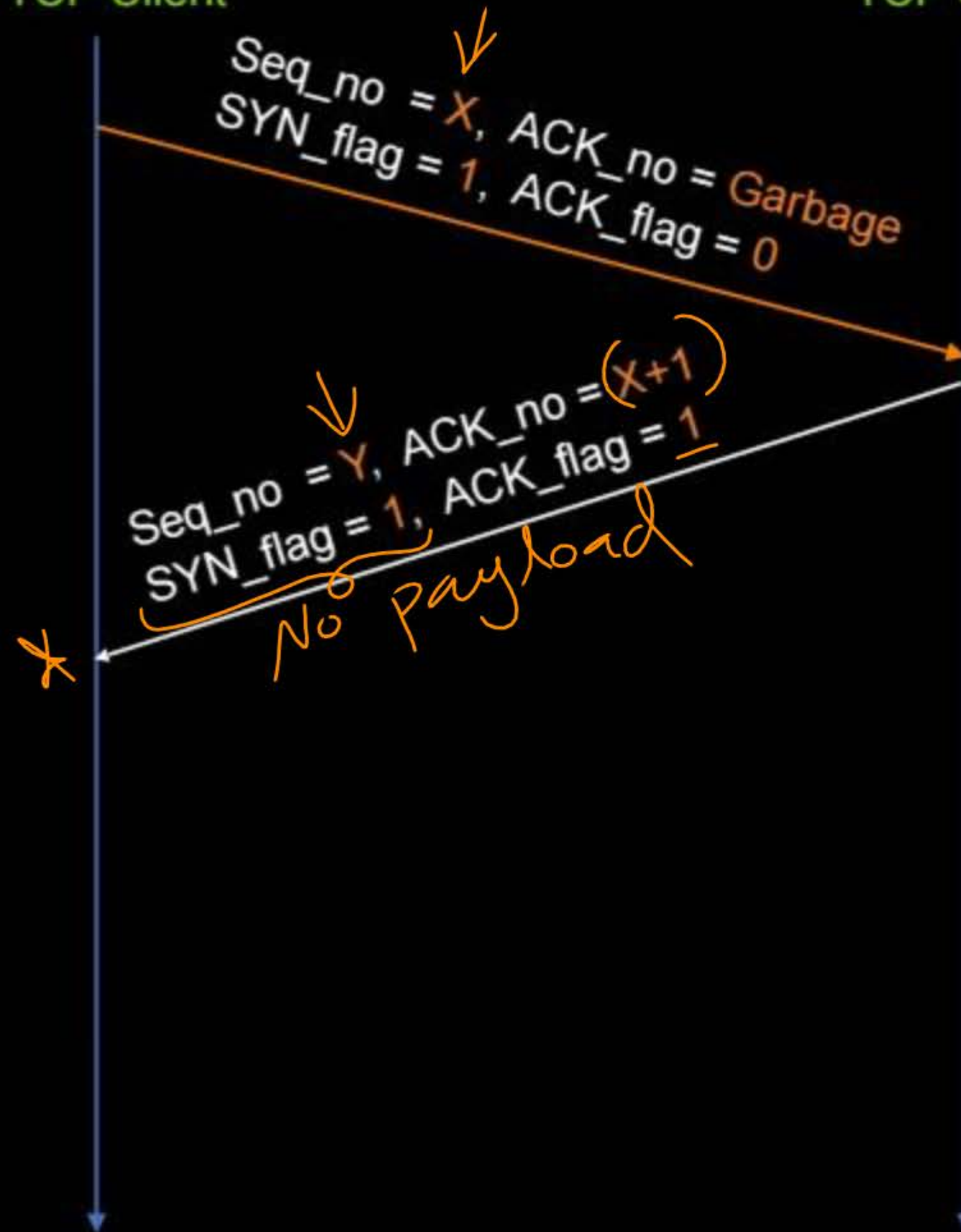


Client_ISN = X

TCP Client

TCP Server

Server_ISN = Y





Client_ISN = X

TCP Client

TCP Server

Server_ISN = Y

Seq_no = X , ACK_no = Garbage
SYN_flag = 1 , ACK_flag = 0

Seq_no = Y , ACK_no = $(X+1)$
SYN_flag = 1 , ACK_flag = 1

Seq_no = , ACK_no = $(Y+1)$
SYN_flag = 0 , ACK_flag = 1

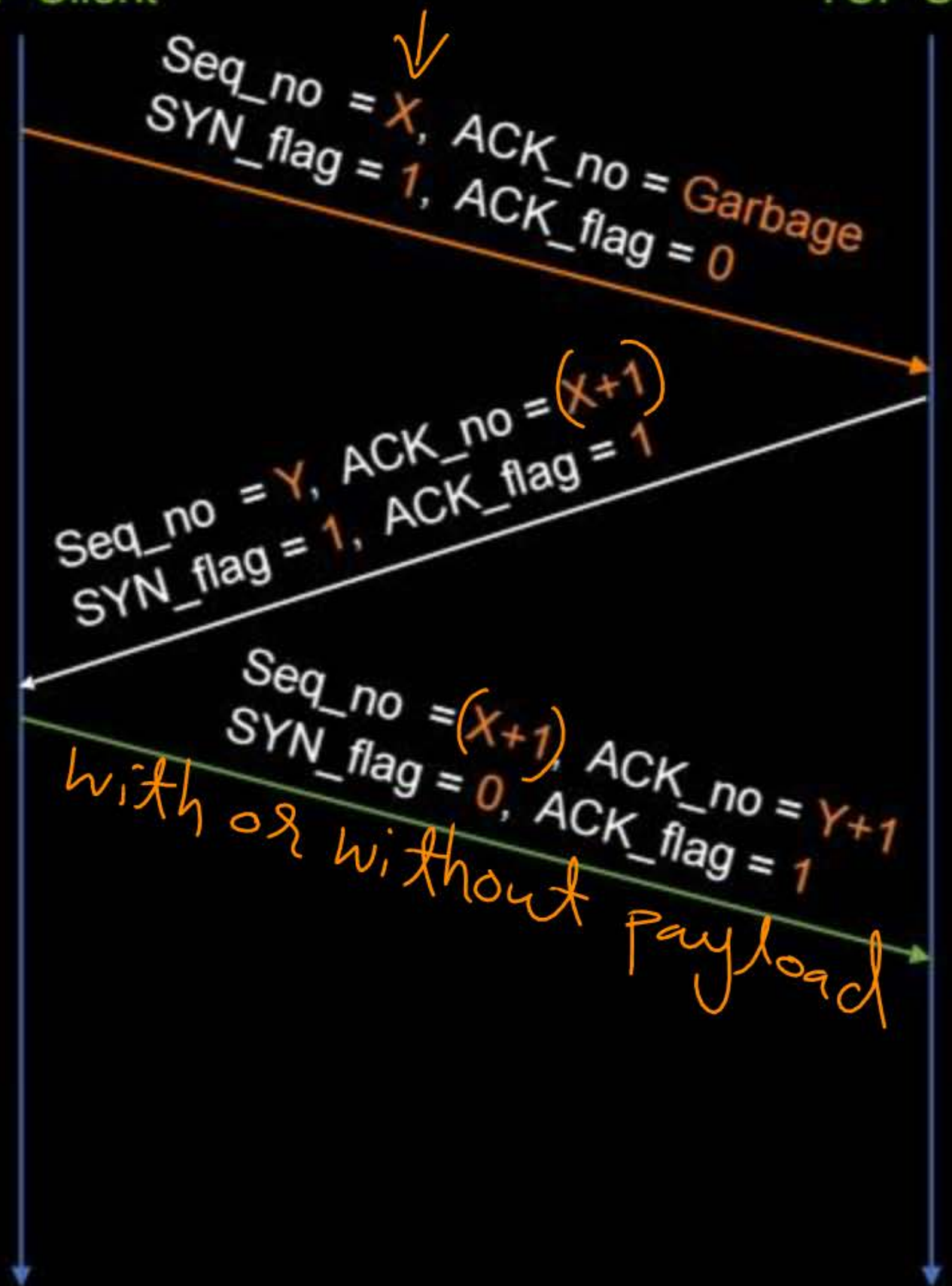


↓
Client_ISN = X
First Byte → (X+1)

TCP Client

TCP Server

Server_ISN = Y



First Byte = (X+1)



Topic : TCP Connection Establishment

→ SYN segment do not carry user data [no any payload]

↓
[syn-flag = 1]



Topic : TCP Connection Establishment



→ SYN segment consume one sequence number

$[syn_flag = 1] \Rightarrow (seq\ no. = ISN)$ →

Ackno. = $(ISN + 1)$ ←

First byte = $(ISN + 1)$ →

[MSQ] [GATE-2024, Set-1] [1-Mark] 115C



#Q. TCP client P successfully establishes a connection to TCP server Q. Let N_P denote the sequence number in the SYN sent from P to Q. Let N_Q denote the acknowledgement number in the SYN ACK from Q to P. Which of the following statements is/are CORRECT?

A The sequence number N_P is always 0 for a new connection False

✓ B The sequence number N_P is chosen randomly by P True

C The acknowledgement number N_Q is equal to N_P False

✓ D The acknowledgement number N_Q is equal to $N_P + 1$ True

Ans: B & D

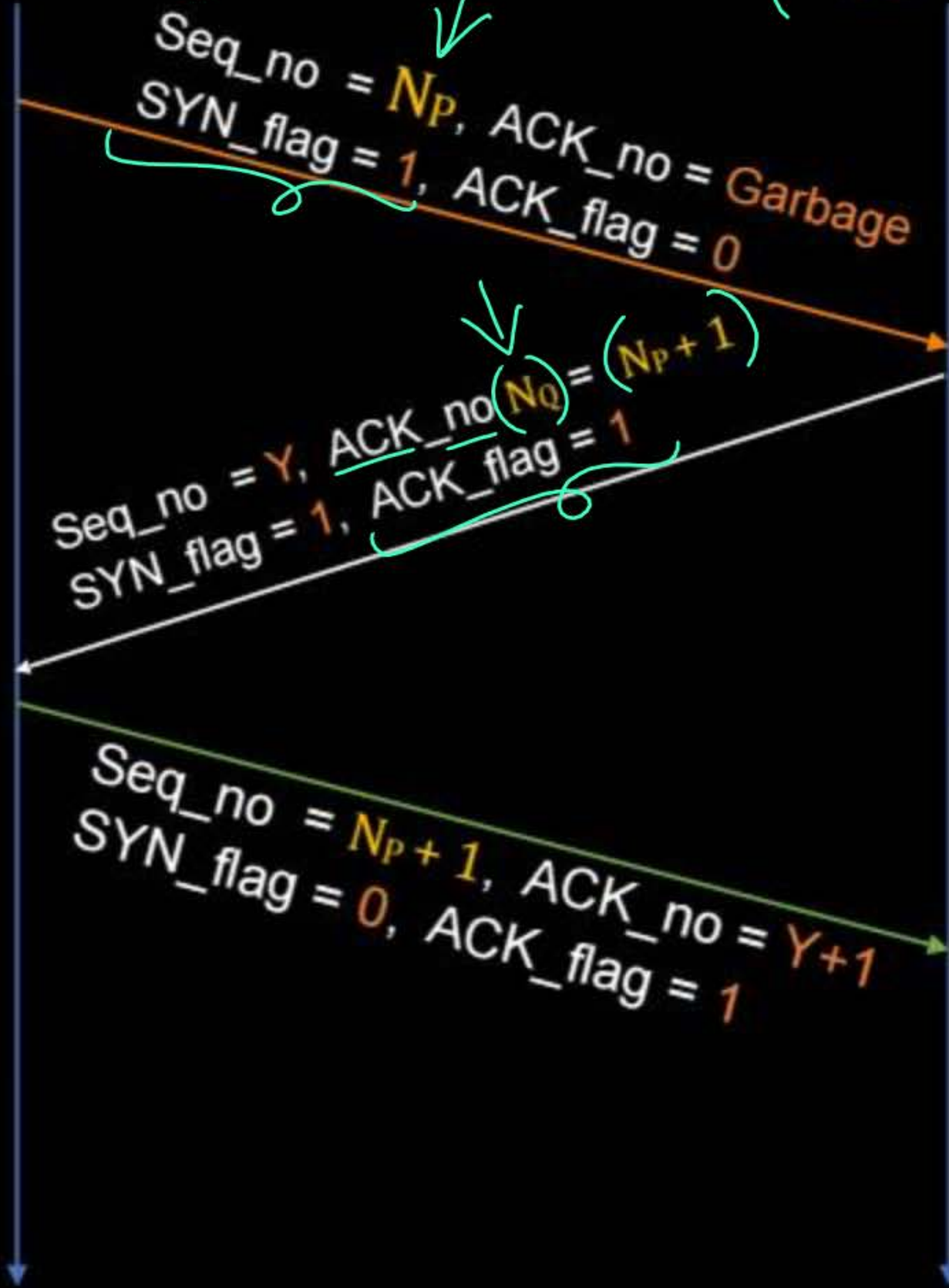


Client_ISN = N_P

TCP Client P

TCP Server Q

Server_ISN = Y





Topic : TCP State Transition

→ Life-cycle of a TCP connection

[Changes in a TCP connection based on the exchange of ~~data packets~~]
TCP segments

TCP Client (P₁) (P₂) TCP Server

CLOSED

SYN_SENT

ESTABLISHED

LISTEN

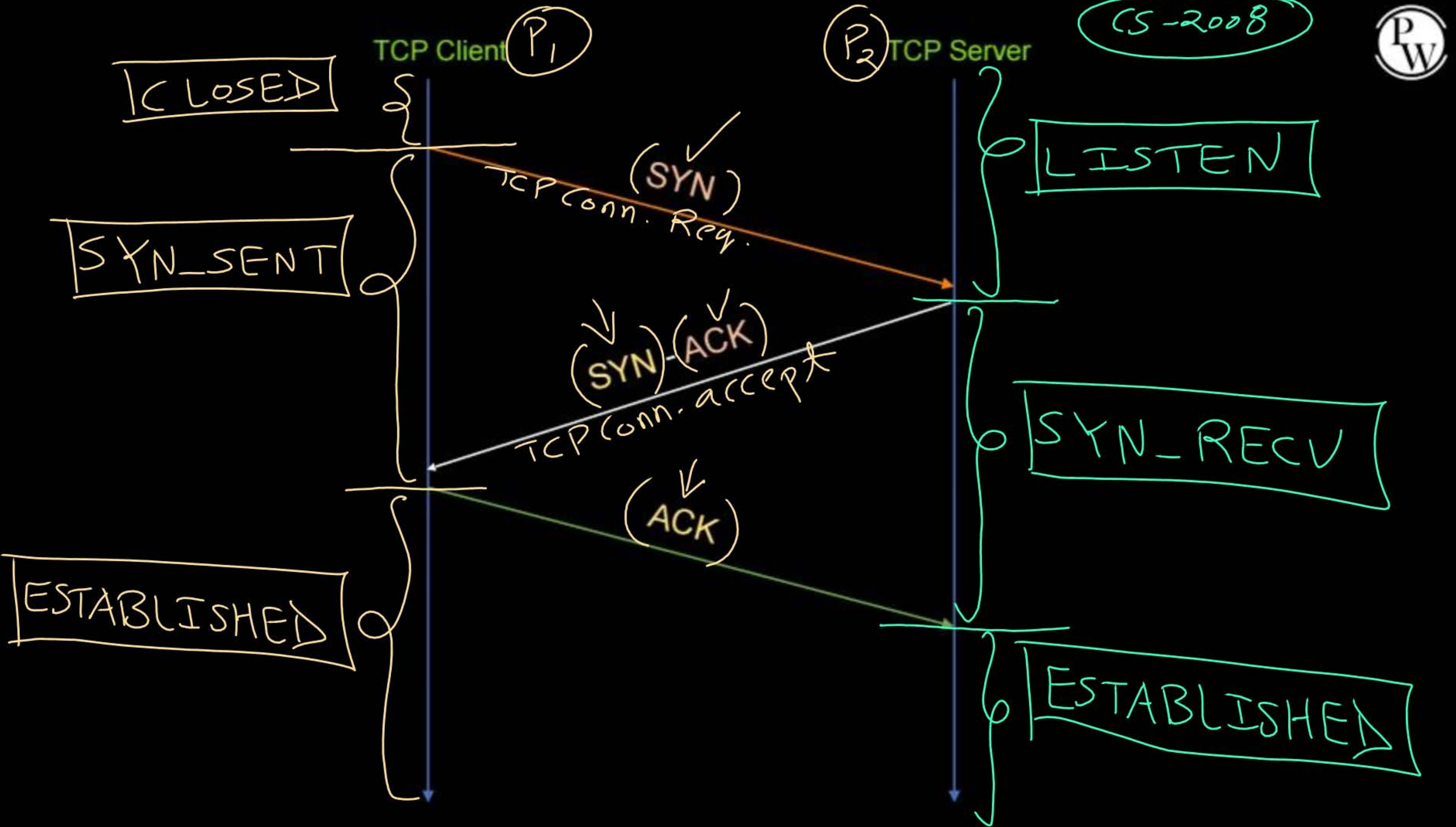
SYN_RECV

ESTABLISHED

(SYN) ✓
TCP Conn. Req.

(SYN) (ACK) ✓
TCP Conn. accept

(ACK) ✓





Topic : TCP



During TCP Data Transfer

→ Full-duplex and point-to-point logical connection

CS-15
CS-26

logical linked

Two TCP connection state





Topic : TCP Connection Close



→ Connection close between TCP client and TCP server



Topic : TCP Connection Close



→ TCP client or TCP server any one can initiate the connection close request

(TCP A & B)

↓
who finish data transmission first;



Topic : TCP Connection Close



→ 4-way handshake process

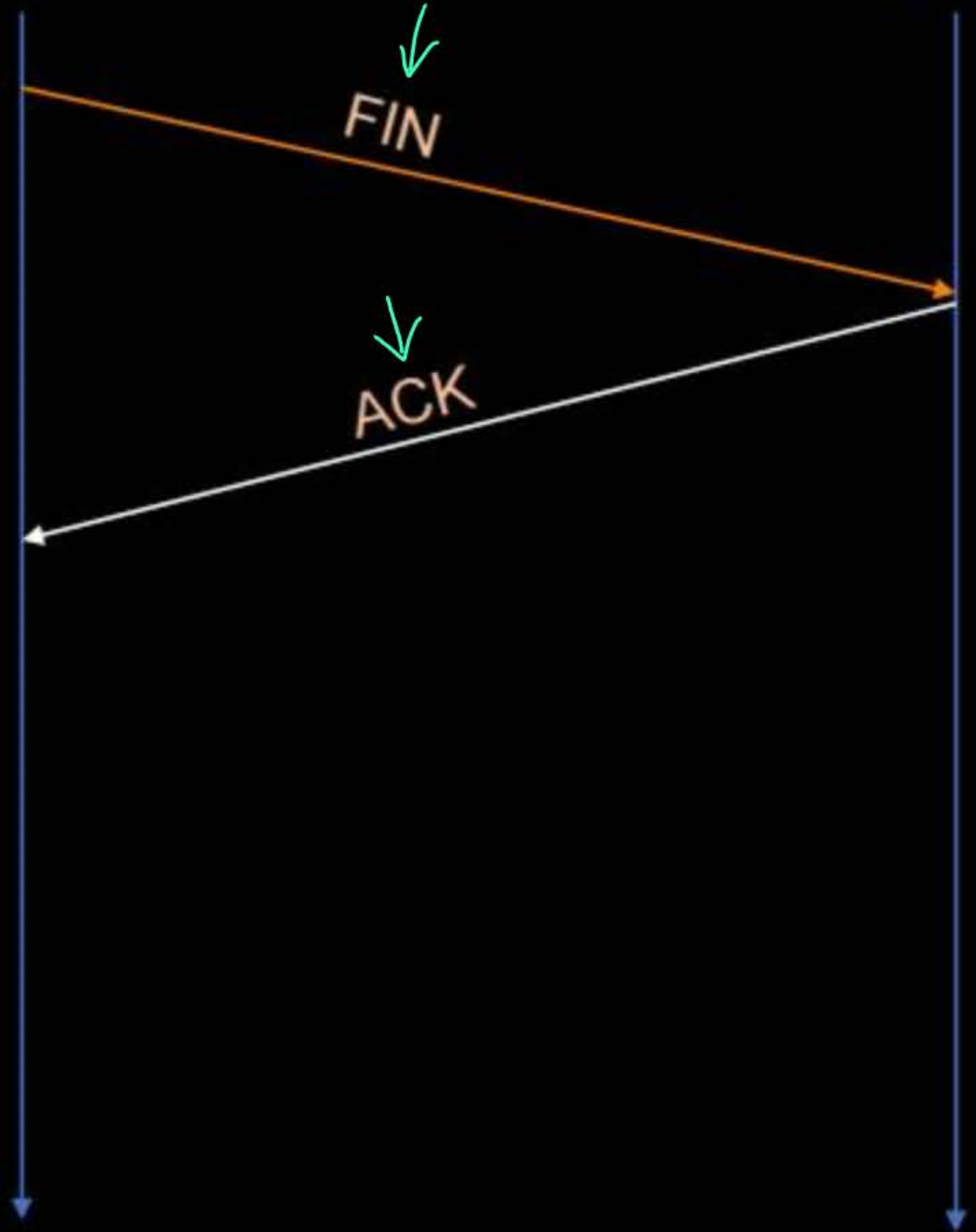
TCP A

TCP B



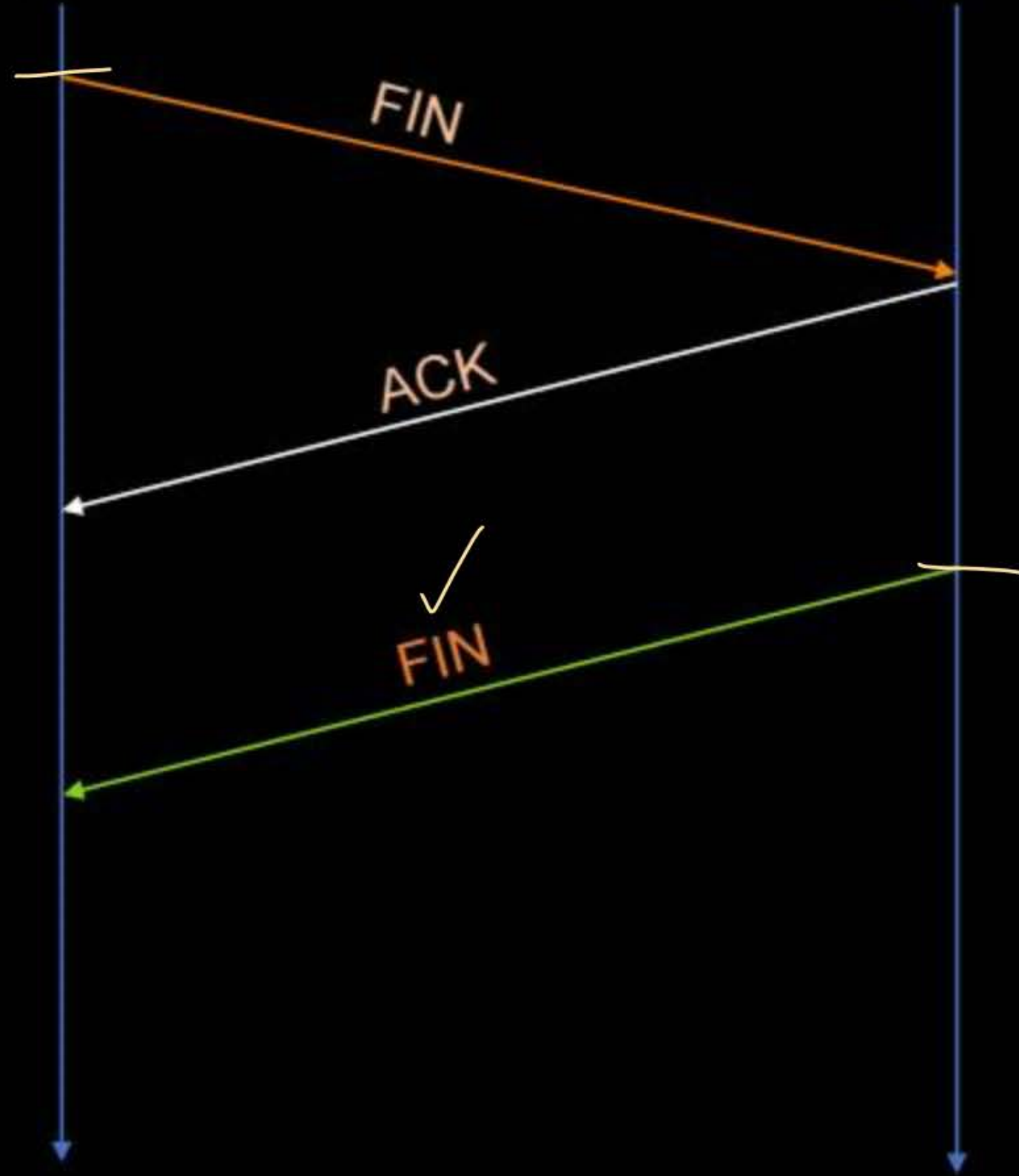
TCP A

TCP B



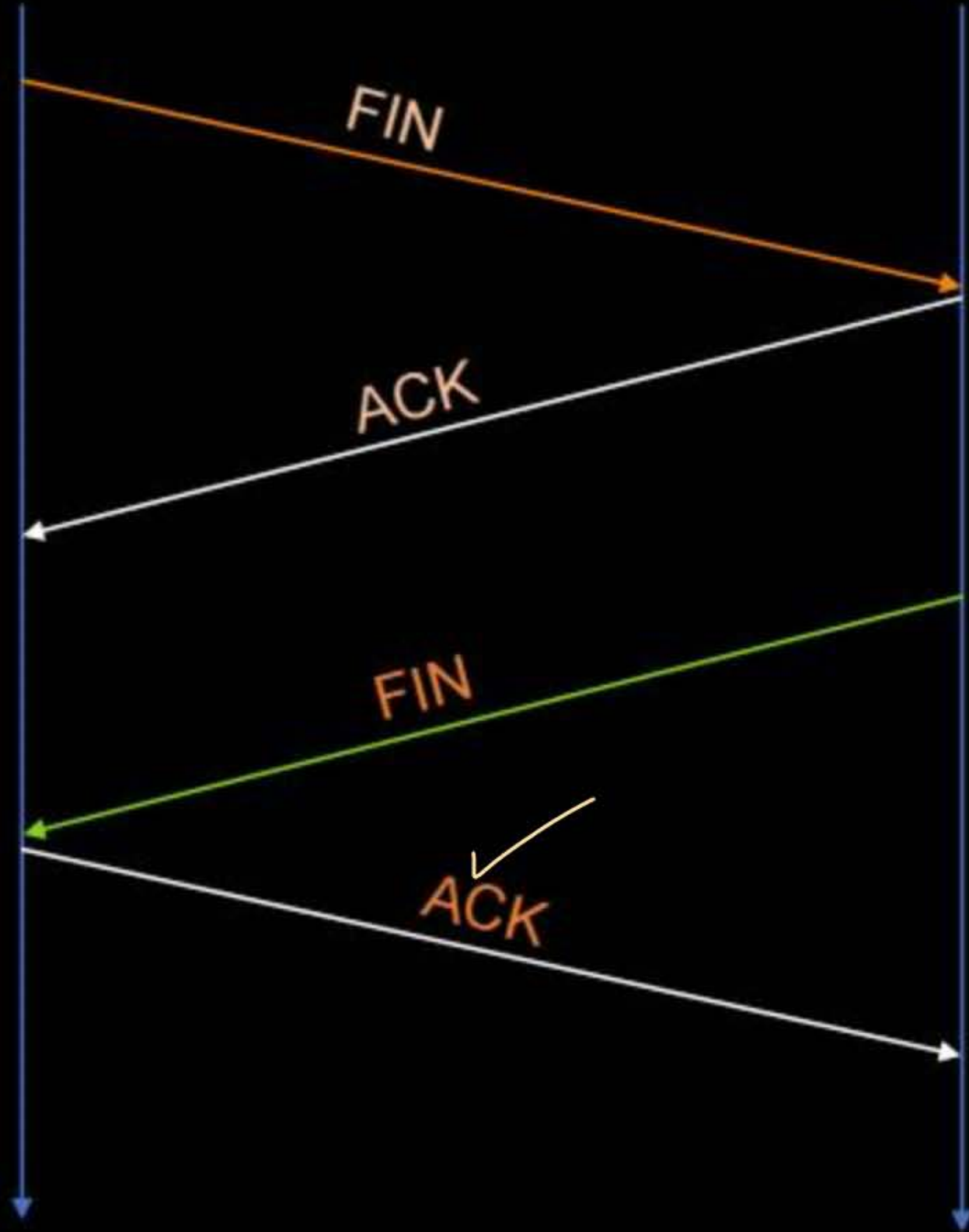
TCP A

TCP B



TCP A

TCP B



TCP A

TCP B

FIN_flag = 1, ACK_flag = 1
with or without payload

TCP A

TCP B

FIN_flag = 1, ACK_flag = 1

FIN_flag = 0, ACK_flag = 1

with or without payload

TCP A

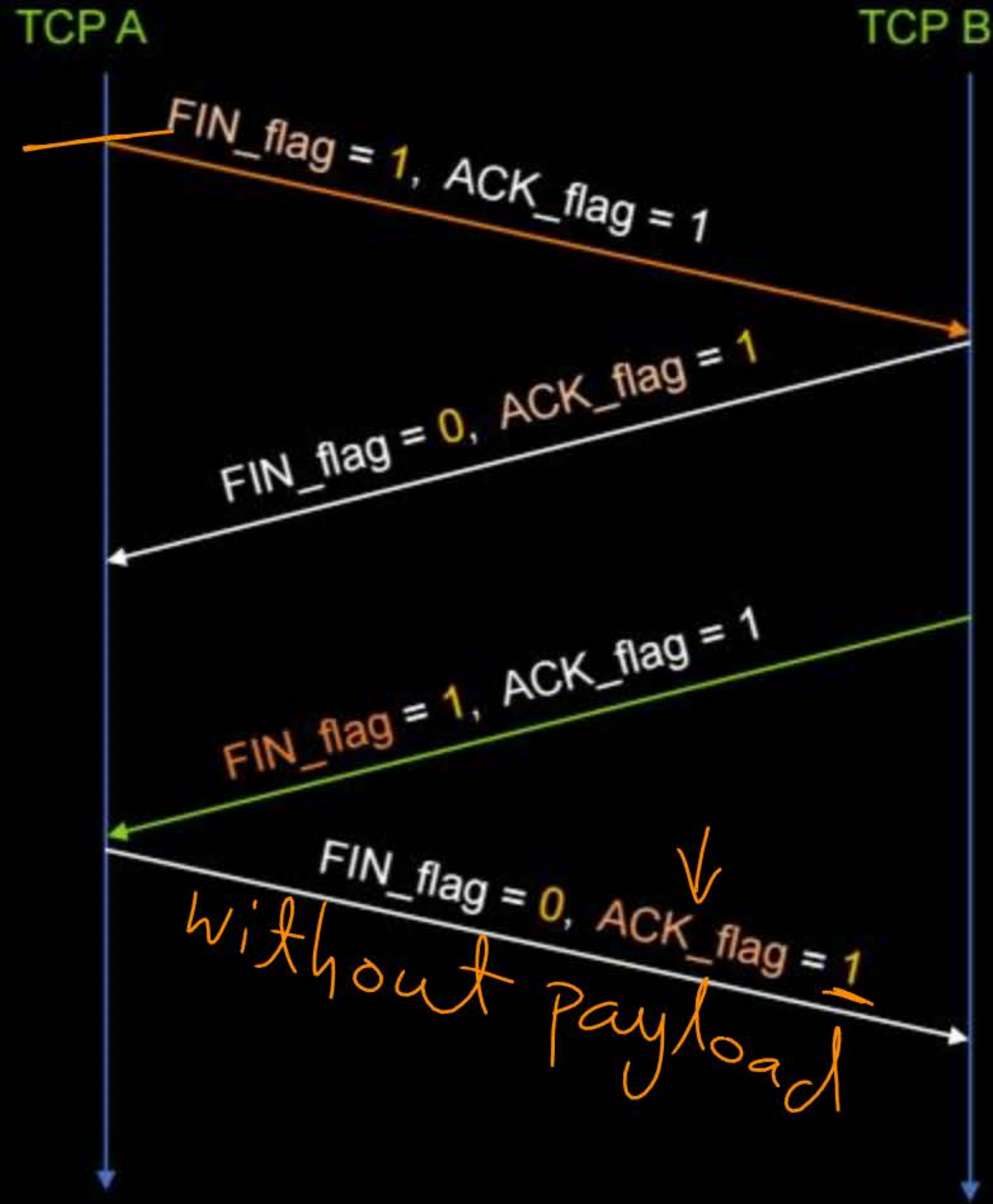
TCP B

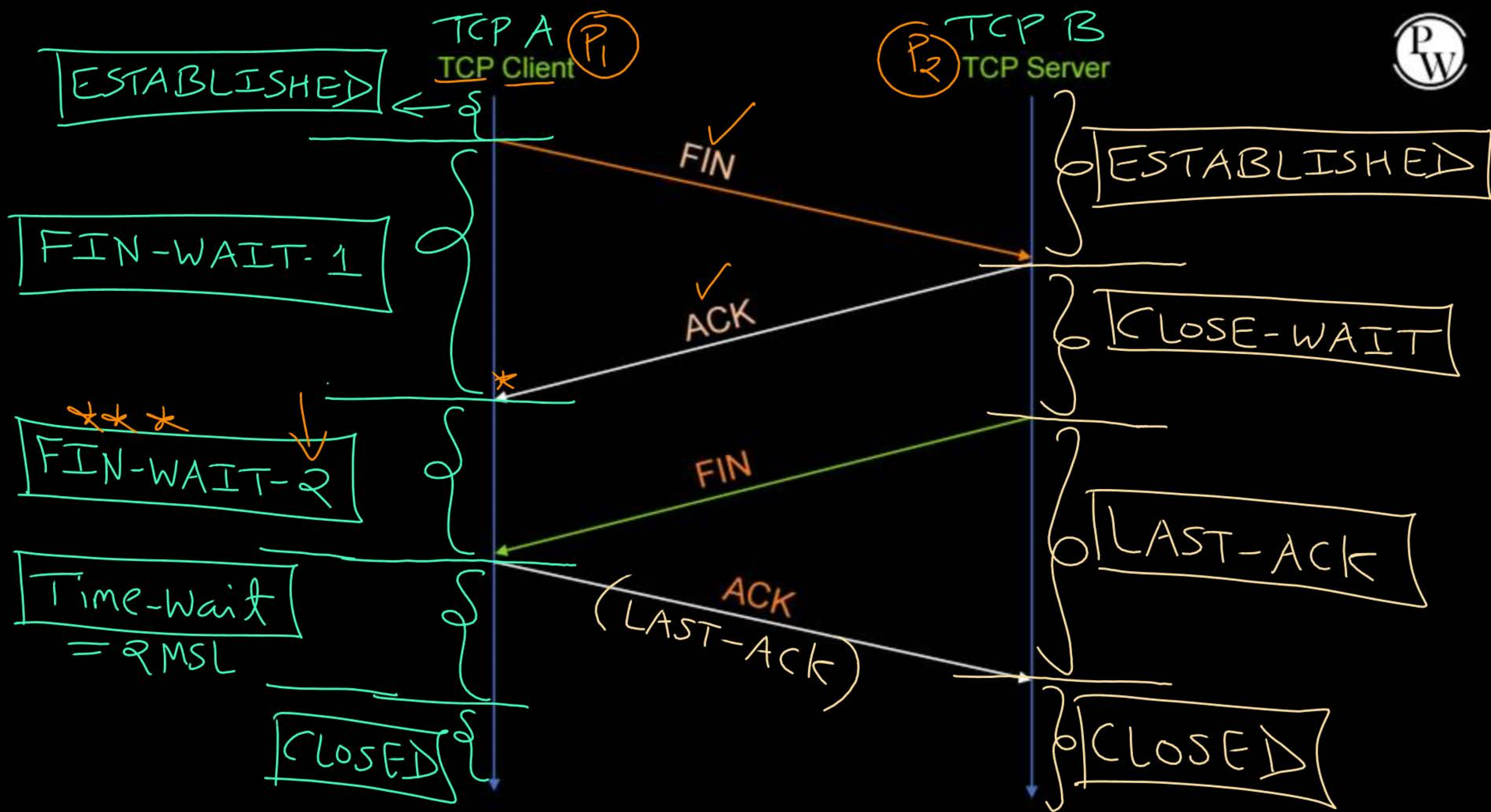
FIN_flag = 1, ACK_flag = 1

FIN_flag = 0, ACK_flag = 1

FIN_flag = 1, ACK_flag = 1

with or without payload





[MCQ] [GATE-2017, Set-1] [1-Mark] (IIT-R)



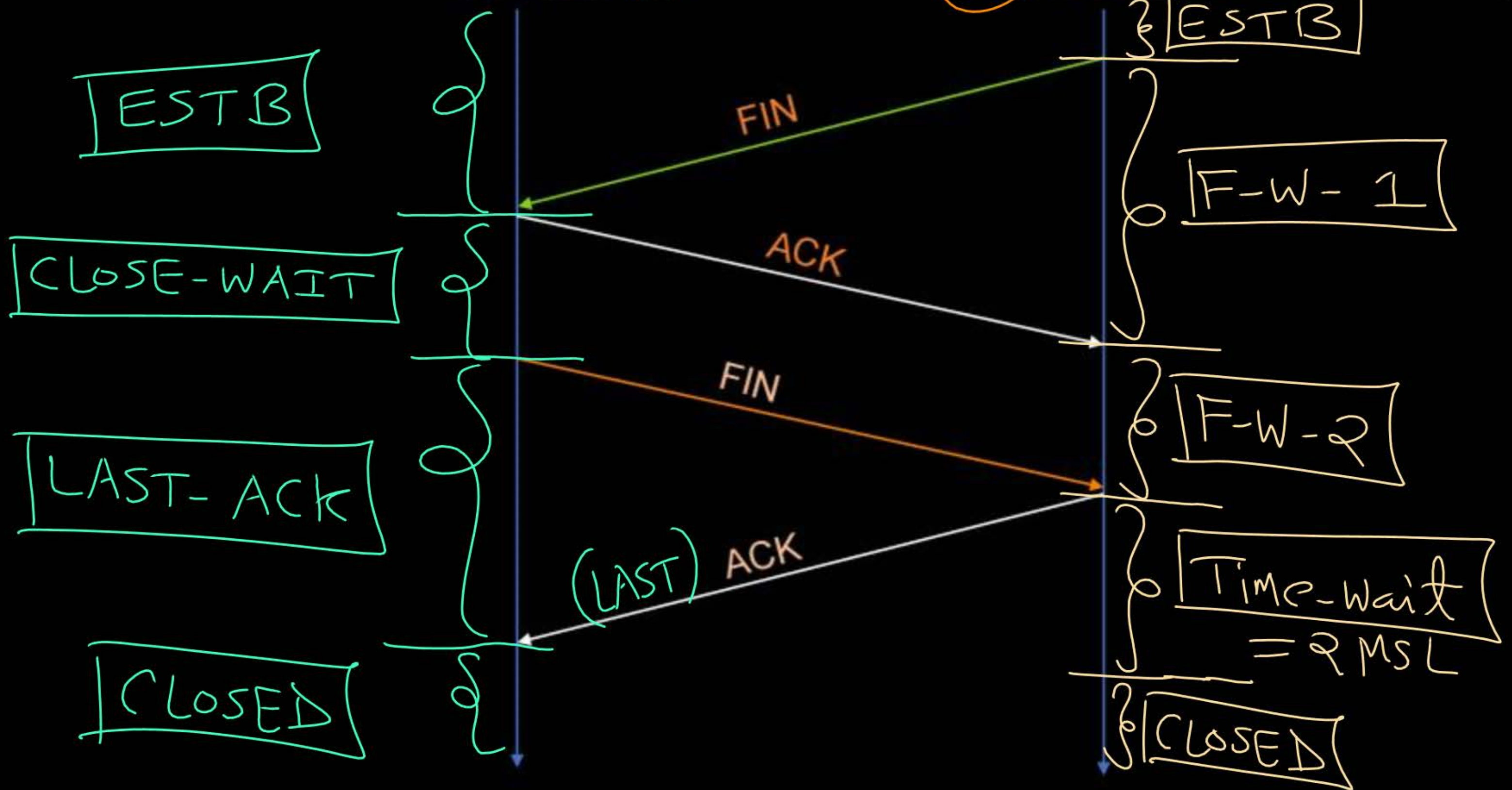
#Q. Consider a TCP client and a TCP server running on two different machines. After completing data transfer, the TCP client calls `close` to terminate the connection and a FIN segment is sent to the TCP server. Server-side TCP responds by sending an ACK which is received by the client-side TCP. As per the TCP connection state diagram (RFC 793), in which state does the client side TCP connection wait for the FIN from the server-side TCP?

- ☐ A LAST-ACK
- ☐ B TIME-WAIT
- ☐ C FIN-WAIT-1
- ☒ D FIN-WAIT-2

Ans: D

TCP B (P₁)
TCP Client

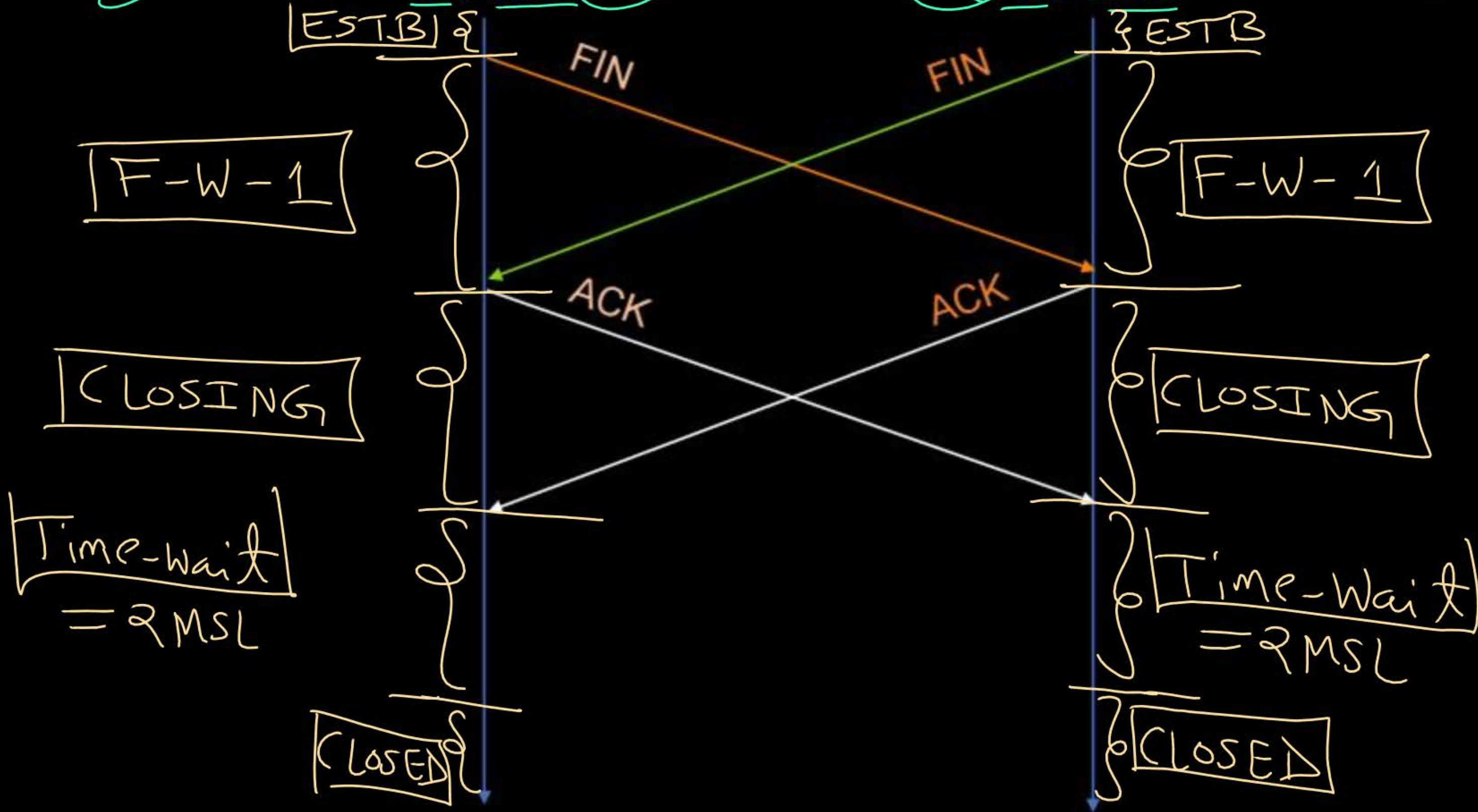
(P₂) TCP A
TCP Server



✓ ✓ ✓ ✓
TCP Simultaneous Connection Close :-

TCP A
TCP Client (P₁)

(P₂) TCP B
TCP Server



#Q. With respect to a TCP connection between a client and a server, which one of the following statements is true?

- (A) The client and server use a two-way handshake mechanism before the start of data transmission False
- (B) The TCP connection is half-duplex False (Full)
- (C) The server ^{can} cannot initiate closing of the connection before the client initiates closing of the connection False
- ✓ (D) The client and server can initiate closing of the connection at the same time True

Ans: D

TCP A

TCP B

ESTB

F-W-1

Time-wait
= 2MSL

Closed

ESTB

(close-wait)

LAST-ACK

CLOSED

FIN

FIN-ACK

ACK



2 mins Summary



Topic

TCP Connection Close

THANK - YOU